RespiraTwin: System Architecture

Multimodal respiratory twin for simulated 24-hour asthma exacerbation-risk forecasting

SYNTHETIC-DATA PROOF OF CONCEPT

1 MULTIMODAL INPUTS

Synthetic historical EHR

Demographics, severity, prior events, comorbidities and controller Rx
Baseline respiratory rate / SpO_2

Synthetic pollution series

PM2.5 / PM10 and source provenance
AQI labelled by its index definition
Illustrative series, not measurements

Wearable + patient diary

Respiratory rate, HR, HRV RMSSD, SpO_2
Sleep/activity, adherence, rescue puffs
Symptoms and simulated timestamps

2 VALIDATION + FUSION

Schema / range validation

Missingness and stale-channel flags
Stale threshold: >6 h

52 past-looking features

Rolling 6 h / 24 h / 72 h summaries
Only observations available at time t
No look-ahead into the forecast window

Personalized baseline

14–30 day lookback window
Excludes the latest 72 hours
Recent signals compared with baseline

3 DIGITAL TWIN ENGINE

Virtual patient state

Profile, telemetry buffers, quality flags
Timestamped state and trajectory

Risk predictor

models/model.joblib (2.15 MB)
Constrained HistGradientBoosting + isotonic probability calibration
24 h simulated risk + 0–100 demo score

Explainability + history

Five signal families + reason codes
Personal-baseline occlusion
Append-only forecast / shift history

4 CLINICIAN DASHBOARD

Patient overview

Selected patient + synthetic-data banner
Historical profile and prior events

Synchronized signal charts

Pollution, wearable, adherence, symptoms
Timestamps and provenance indicators

Risk forecast + explanations

Estimated 24 h simulated risk
Score, demo band and contributing signals

Twin history + shift audit

Timestamped trajectory and prior forecasts
Inspect state changes over time

What-if laboratory

Isolated scenario copy; recompute features
Hypothetical inputs only
Original observations and history unchanged

Early-warning replay

Replay held-out synthetic episodes
Inspect anticipated, late and missed events
Evaluation based on simulator labels

What-if scenario reuses feature pipeline; stored history unchanged

Trained artifact loaded by predictor at runtime

OFFLINE TRAINING AND EVALUATION

Synthetic cohort + labels

Generate patient profiles and hourly streams
Simulated event onsets / labels
No real patient data

Temporal features + patient split

Past-looking feature construction
Train / validation / test by patient
210 / 45 / 45 patients

Model comparison + calibration

Logistic Regression / Random Forest / HGB
Constrained HGB selected
Isotonic calibration on validation only

Artifacts

model.joblib (2.15 MB)
metrics.json • whatif_effects.json
Loaded by the runtime predictor above

SAFETY NOTE Synthetic-data proof of concept only; results do not establish real-world clinical accuracy.